

摘 要：再生混凝土由废弃混凝土经破碎、筛分后作为骨料重新拌制而成，可同时缓解建筑固废堆存压力与天然砂石资源紧张。本文以广西某 30 层、建筑高度接近 100 m 的再生混凝土公寓项目为对象，对该项目中再生骨料取代率为 30%、总用量约 10679 m³ 的再生混凝土应用情况进行评价。研究从技术性能、施工适应性、经济与环境效益三个维度构建评价指标体系，结合配合比设计、构件生产与实体检测数据开展分析。结果表明：30%取代率条件下再生混凝土的抗压强度、弹性模量与耐久性能均满足设计要求，弹性模量略低的特点有利于优化结构刚度分布、改善地震作用下的内力分配；但骨料吸水率偏高导致的工作性波动、构件质量离散性偏大等问题仍需控制。对此提出加强骨料分级预处理、优化配合比参数、完善过程质量留痕等对策。

关键词：再生混凝土；装配式建筑；住宅项目；应用评价；再生骨料

Abstract: Recycled aggregate concrete is produced by crushing and screening waste concrete and reusing it as aggregate, which can ease both the pressure of construction waste storage and the shortage of natural sand and gravel resources. Taking a 30-storey apartment project with a height of nearly 100 m in Guangxi as the research object, this paper evaluates the application of recycled aggregate concrete with a replacement ratio of 30% and a total dosage of about 10679 m³. An evaluation index system is established from three dimensions: technical performance, construction adaptability, and economic and environmental benefits. Combined with the mix proportion design, component production and entity inspection data, the applicability of recycled concrete in high-rise shear wall structures is analysed. The results show that under the condition of 30% replacement ratio, the compressive strength, elastic modulus and durability of recycled aggregate concrete can meet the design requirements, and the slightly lower elastic modulus helps to optimize the structural stiffness distribution and improve the internal force distribution under earthquake action. However, problems such as workability fluctuation caused by the high water absorption of recycled aggregate and the large discreteness of component quality still need to be controlled. Countermeasures including strengthening the classification and pretreatment of recycled aggregate, optimizing mix proportion parameters and improving the whole-process quality records are put forward.

Key words: recycled aggregate concrete; prefabricated building; residential project; application evaluation; recycled aggregate

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